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Blockchain for businesses: The ultimate enterprise guide

TIP

Top 9 smart contract platforms to consider in 2025

Platform speed, scalability, security and smart contract development support vary widely. Here's how to match features to your needs plus the lowdown on the major players.

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Smart contracts let users write executable code that encodes business logic and runs on a blockchain or other type of decentralized ledger. Depending on your goals, there are many ways to use smart contract platforms to push these powerful apps into production.

"Smart contract platforms are decentralized systems that enable self-executing contracts on a blockchain," said Sanjiv Maewall, CTO of [blockchain](#) and digital assets at Deloitte Consulting. These platforms support secure, transparent and tamper-proof execution of predefined conditions, according to Maewall, which makes them ideal for trustless applications such as finance, supply chain management and voting systems.

There are four main kinds of smart contract platforms: large [blockchain platforms](#); smaller ones optimized for improving smart contract execution speed or cost; private

smart contract frameworks; and specialized tools for crafting and managing smart contracts across trusted partners.

These tools were selected based on their prominence across the various smart contract platforms; conversations with industry experts; and, in the case of the public smart contract platforms, their market cap and trading volume.

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— [Top 10 benefits of blockchain for business](#)

— [What are the 4 different types of blockchain technology?](#)

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What are smart contract platforms, and how do they work?

Before diving deeper into the different types of smart contract platforms, it's helpful to define what a [smart contract](#) is and what it does.

Lata Varghese, digital assets and blockchain practice leader at Protiviti, said she likes to think of a smart contract as "a digital form of a traditional contract where contractual agreements between counterparties are inscribed as a piece of code." A smart contract is a self-executing program that automates the actions required by an agreement when predetermined terms and conditions are met. It ensures trust between the involved parties without needing a central authority. A smart contract platform is a blockchain-based decentralized platform that lets users create and te smart contracts.

A blockchain is not in and of itself a smart contract platform. Developers must extend it with tools to store smart contract code in a secure and tamper-proof form. They also add tools that enable decentralized and autonomous execution of smart contract code correctly, securely, transparently and deterministically. It's also important for the state transition from smart contract code execution to be verified by consensus of the nodes in the blockchain network before recording the state change in the blockchain ledger.

The blockchain-based cryptocurrency [Bitcoin](#) supported the first digital, permissionless and tamper-proof smart contract, but it was of limited scope. Varghese said the major breakthrough came when the developers of the Ethereum blockchain added support for programmable smart contracts. They also introduced the Ethereum Virtual Machine (EVM) as the computation engine that manages the state of the blockchain and enables smart contract functionality.

One caution for enterprise users is that smart contracts are not legally binding. Within the Web3 community, the term is used differently than enterprises looking to automate contract processes in a traditional sense. In parallel, lawyers and governments are developing frameworks for what are called smart legal contracts, which might or might not use decentralized ledger technology. Automated procurement platforms, such as SAP Ariba and B2B marketplace platforms, can sometimes be used to automate contractual processes through ERP integrations. Contract lifecycle management platforms ([CLMs](#)), such as the DocuSign Agreement Cloud, are also exploring ways to integrate smart legal contracts into enterprise workflows.



What is a Smart Contract and What is it Used For?



Layers and levels of smart contract platforms

Several [blockchain technology](#) platforms have been used for executing smart contracts. They're usually characterized as Level 1 (L1) or Level 2 (L2) platforms.

L1 platforms are blockchains on which smart contracts are executed directly on the main core platform. Prominent examples are Ethereum, Solana, Cardano and BNB Smart Chain.

L2 platforms execute the smart contracts on a parallel space so they can run more

platforms include Ethereum ZK-Rollups, Ethereum Optimistic Rollups and Polygon. Many of the newer offerings support both L1 and L2 capabilities.

"There is continuous innovation occurring in these technology platforms to address blockchain scalability challenges and become the platform of choice to build Web 3.0 applications," Varghese said, referring to the future blockchain-based generation of the web, often called [Web3](#).

The newer blockchain platforms almost uniformly have support for smart contracts, she said, and many are EVM compatible. That means a decentralized application (dApp) that runs on Ethereum can also run on them. Other prominent platforms

Varghese has explored include Avalanche, Tezos, Algorand, Polygon, Cosmos, Polkadot, Tron, Near and Cardano.

In addition, some blockchain platforms, such as Avalanche, Cosmos and Polkadot, are being adapted for private, permissioned smart contract implementations. Several smart contract platforms, such as [Hyperledger](#) Fabric, R3 Corda and Hyperledger Besu have supported enterprise blockchain implementations for at least five years.

 A list of major smart contract vulnerabilities.



Key factors to consider in a smart contract platform

When choosing a smart contract platform, Maewall suggested examining a platform's [security features](#), track record, scalability potential, performance, cost and strength of the associated developer community. "Each platform has its unique offerings and tradeoffs, so choose one that best aligns with your project's requirements," he said.



Purchasers should investigate a platform's protections against these smart contract security risks.

These platform elements also come with compromises. For example, Ethereum has the strongest developer community, but it also has the

highest cost per transaction -- called *gas* in the blockchain community -- Maewall said. Similarly, some platforms are Turing complete. They can simulate every function of a Turing machine, a mathematical model of computation, which means users can write more expressive applications. But this flexibility comes at the expense of [security vulnerabilities](#) that must be addressed.

 Understanding the different approaches

In some ways, smart contract infrastructure is comparable to that of cloud computing, which has many nested and sometimes overlapping platforms. On one level are the large infrastructure platforms such as [AWS](#), Microsoft Azure and Google Cloud. The application software layer includes Salesforce, ServiceNow and other platforms. In addition, private clouds enable enterprises to deploy workloads running on [Docker](#) containers that can move across public and private cloud infrastructure.

Similarly, in the smart contract arena, Ethereum is like AWS before Microsoft and Google came along. Several newer competitors, including Cardano, Solana and Tron, provide alternatives to Ethereum. Another tier of platforms that includes Hyperledger Fabric, Corda and Tezos mimics [private cloud](#) infrastructure. Development management platforms, such as Kaleido and Casper Labs, are designed to make it easier for partners to create and manage their own smart contract infrastructure. Additionally, CLM platform vendors are extending their tooling to support smart legal contract development management capabilities that span decentralized ledger and CLM capabilities, such as DocuSign Smart Agreements.

There are nuances in how the various platforms approach the problem of smart contract development and deployment. For example, apps might run on EVMs, low-level VMs, Docker containers or other infrastructure.

It's also important to consider your reasons for building decentralized apps before choosing a smart contract platform. Differences in platform features and capabilities can determine what an app can do and what users can accomplish by using it. For example, platforms such as Ethereum and Solana make it easier to incentivize infrastructure creation with cryptocurrency tokens. However, private trading partners might be happy to build or rent their own infrastructure. They are more motivated to streamline transactions and improve transparency.

Much like in the cloud industry, there are dozens of smart contract platforms, and the field is evolving rapidly. Rather than surveying all of them, it's more helpful to focus on the top categories and the biggest or most interesting players. With this in mind, let's look at some of the top smart contract platforms across key categories.



Ethereum pioneered flexible smart contracts when it was launched in 2013. It supports Solidity, a variant of the [JavaScript](#) programming language that was immediately accessible to front-end developers and quickly grew a massive base of developers and financial supporters. Ethereum also pioneered the EVM, which has become a standard for other smart contract platforms.

Ethereum's stock market capitalization is far and wide above competitors. The main benefit of a large market cap is that successful smart contract apps have the potential to grow faster in a much larger virtual economy. Ethereum is the top choice for developers who want to game the largest [smart contract ecosystem](#) with a bright new idea.

Some observers say Ethereum's performance is not as fast as some of the upstarts. Smart contract mistakes can be expensive and difficult to unwind; in some cases, this can require a *fork*, a usually temporary split in the Ethereum network to allow major technical upgrades or changes. The blockchain has also been undergoing a constant pace of forks, merges and upgrades. The Shanghai upgrade in mid-2023, called Shapella, supported a more efficient proof-of-stake consensus mechanism and faster transaction processing.

Second generation

The success of Ethereum smart contracts revealed some fundamental disadvantages of the platform. For one thing, it's not particularly fast compared to modern transaction processing infrastructure -- a few dozen transactions per second compared to 24,000 for the Visa payment network.

Hackers also discovered new security bugs in Ethereum smart contracts that were difficult to resolve. For example, a hack of the first Ethereum distributed [autonomous organization](#), a type of management structure built on a smart contract, resulted in the DAO losing a third of its value. Eventually, Ethereum was hard forked -- permanently split in two -- to undo some of the damage. A second generation of open blockchain networks has sprung up to address such shortcomings.

BNB Smart Chain

BNB Smart Chain is built on the BNB coin. It was previously the Binance Coin by Binance and was migrated to a separate Smart Chain in 2019 to create a native coin.

The platform supports a proof-of-stake authority consensus mechanism architected to optimize speed and keep transaction costs low. The network currently supports a million daily users and more than 2,000 active DApps. It is also popular among the gaming community, with over 287,000 weekly active gamers. The platform supports several popular development languages, including Solidity, Go, Java and Swift.

One potential concern for developers is that Binance, the company behind the chain, has suffered legal problems that could impact the rollout of smart contract apps in the U.S. In 2023, Binance [pleaded guilty](#) in U.S. federal court to failing to maintain an effective anti-money laundering program and agreed to pay over \$4 billion in fines.

Solana

The Solana blockchain ecosystem has grown to be the third largest public smart contract platform. Some experts consider it more efficient than Ethereum and many other public blockchains. Factors contributing to the speed include support for a low-level virtual machine for deploying smart contracts as well as the use of [C](#) and Rust [programming languages for writing them](#). That makes it a good bet for developers looking to combine high transaction rates with the economic opportunities available on a public blockchain.

Cardano

Cardano was introduced as a more energy-efficient alternative to Ethereum in 2015 by Charles Hoskinson, who also helped start Ethereum. In 2021, it introduced its Plutus feature for building smart contracts. The community has also created several programming languages for creating smart contracts on Cardano, including Marlowe, Aiken, OpShin and plu-ts. Cardano is a newer platform with less functionality and fewer developers than others on this list, but it is an interesting option for developers who want to take advantage of a large token economy outside of Ethereum.

Private blockchains

Private blockchains are similar to open source cloud development platforms, such as Cloud Foundry, OpenShift and Cloudify but for smart contracts. The code itself is open source; a variety of expert service firms provision it on private hardware in data centers or on top of cloud services.



Private blockchains are a good fit for businesses and government organizations that want to streamline transactions between trusted partners. They're a good choice for improving the efficiency of things enterprises already do, with mechanisms to update contracts or roll back transactions when problems are discovered. The most common use of private blockchains is scaling trust across enterprise boundaries.

Hyperledger Fabric

IBM developed Hyperledger Fabric and then handed it off to the [Linux](#) Foundation to create an enterprise-grade blockchain platform with modular architecture and various consensus mechanisms. It enables fine-grained control of data visibility and confidentiality so only authorized people can access data. It also supports multiple programming languages, including JavaScript, Go and Java, and is designed to handle high transaction volumes.

Hyperledger Fabric is a [permissioned blockchain](#) network. That means only authorized individuals or devices can participate in transactions, which is important from a trust, compliance and regulatory perspective.

Corda

Corda was created to model and automate real-world transactions in a legally enforceable way. Private channels let participants transact directly to improve the confidentiality of transactions. Numerous integrations have been built for the finance, healthcare and [supply chain](#) industries.

Smart contract management platforms

Smart contract development and management platforms have emerged to help manage smart contract infrastructure across various platforms. These tools are a good choice for enterprises that want flexibility and adaptability in scaling trust across enterprise boundaries the way Salesforce and ServiceNow created platforms for building business functionality across cloud platforms.

Casper Labs

The Casper platform helps developers write smart contracts using their programming language of choice and deploy them on the Casper blockchain, which lets them execute in a decentralized and secure manner. The platform includes [software](#)

[development kits](#), documentation, code samples and libraries to assist developers in creating and testing their smart contracts.

One of the biggest benefits of the Casper platform is its flexibility. It can support hybrid applications that connect private blockchains such as Hyperledger Fabric with Casper's public blockchain to provide the benefits of both types. It is incompatible with the EVM, which the vendor said makes it more interoperable with existing enterprise technology stacks and IT infrastructure. Casper also recently partnered with IBM Consulting to use AI solutions to improve auditability and compliance of smart contracts.

Kaleido

Kaleido provides tools and services to simplify the process of building, testing and deploying smart contracts on various blockchains. The platform offers pre-built smart contract templates that cover common use cases and industries. The templates serve as starting points for developers, enabling them to more quickly create smart contracts tailored to their specific needs and later customize and extend them.

Kaleido also provides tools for testing smart contract functionality that integrate with a variety of smart contract testing frameworks. The platform can also help automate the smart contract deployment process and then monitor to ensure smart contracts perform as intended. The platform also allows creating contracts from templates, code uploads and imports from other blockchains. It also gives developers tools to manage contracts after deployment, such as upgrades in one place that take effect across all contract deployments.

DocuSign Smart Agreements

CLM pioneer DocuSign acquired Clause in 2021. Clause was a pioneer in smart legal contracts that could be integrated into business applications. The initial platform supported applications in logistics and insurance. For example, self-executing contracts could dynamically alter the price of goods in transit if shipping temperatures went out of a defined range. In insurance, parametric contracts could dynamically adjust the insurance cost in response to external weather conditions.

DocuSign has not spelled out many details about smart contract development as a complete toolset. Rather, it is gradually bridging the notions of CLM and smart

agreements through AI capabilities like DocuSign Insight and connectors to popular procurement platforms such as the SAP Ariba cloud. It also focuses on making it easier for business users and legal teams to craft executable contracts rather than developers writing code that requires a different way of thinking than the logic built into legal agreements.

George Lawton is a journalist based in London. Over the last 30 years, he has written more than 3,000 stories about computers, communications, knowledge management, business, health and other areas that interest him.

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